

Emmanuel Kiprotich

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 LinkedIn |  GitHub |  Portfolio

Summary

Software Engineer and Machine Learning Researcher experienced in AI-powered SaaS, computer vision, cloud applications, and reproducible ML systems. Published research in deep reinforcement learning and ongoing work in federated learning for cybersecurity and transportation systems.

Education

University of Iowa

Master of Computer Science, Incoming Student

Iowa City, IA

Fall 2026

Minnesota State University, Mankato

Bachelor of Science in Computer Science; Minor in Mathematics

Mankato, MN

May 2026

Experience

Amazon Web Services

AWS AI & ML Scholar

2026 – Present

Remote

- Completed applied training in generative AI, responsible AI, SageMaker, Hugging Face, and MLOps.
- Built model-deployment workflows using FastAPI, Docker, AWS endpoints, monitoring, and evaluation metrics.
- Applied prompt engineering and cloud architecture principles through hands-on AI labs and projects.

Minnesota State University, Mankato

Machine Learning Researcher – Federated Learning & Intrusion Detection

Jan. 2025 – Present

Mankato, MN

- Developed federated-learning intrusion-detection models for transportation IoT without centralizing sensitive data.
- Evaluated NSL-KDD and Car-Hacking datasets under non-IID settings using classification metrics and ablation studies.
- Investigated client heterogeneity, federated optimization, and communication-efficient training using Python and Pandas.

Minnesota State University, Mankato

Undergraduate Research Assistant – Deep Reinforcement Learning & V2I

Jan. 2025 – May 2026

Mankato, MN

- Built DQN agents and Gym-compatible SUMO environments for adaptive traffic-signal control using V2I data.
- Reduced queue length by 22%, waiting time by 23%, and increased throughput by 11% over fixed-time control.
- Co-authored *Smarter Intersections*, accepted at ACDSA 2026, supporting tuning, analysis, and manuscript preparation.

Selected Projects

Clausly.app – AI Contract Intelligence SaaS | clausly.app

- Built a Next.js, React, TypeScript, Supabase, and PostgreSQL platform for contract analysis and document Q&A.
- Designed a RAG pipeline using PDF parsing, chunking, semantic retrieval, structured prompts, and grounded responses.
- Developed AI workflows for clause extraction, risk analysis, deadlines, reminders, and document portfolios.

USDA Alfalfa Segmentation – Team Lead | [GitHub Organization](https://github.com/USDA-ARS/alfalfa-segmentation)

- Led a five-member team developing YOLO instance-segmentation models for 20x alfalfa microscopy images.
- Managed preprocessing, annotation review for 369 images, augmentation, hyperparameter tuning, and evaluation.
- Coordinated USDA and University of Minnesota stakeholders, Git workflows, documentation, and Gradio integration.

Wrex.app – AI Writing Intelligence Platform | wrex.app

- Built and deployed an AI writing platform using React, TypeScript, FastAPI, Supabase, PostgreSQL, and GPT-4o mini.
- Developed NLP workflows for authenticity scoring, voice consistency, grammar, rewriting, and rubric alignment.
- Implemented JWT authentication, Stripe billing, rate limiting, testing, and deployments on Vercel and Render.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C, R

AI/ML: PyTorch, scikit-learn, YOLO, OpenCV, Hugging Face, RAG, Computer Vision, Reinforcement Learning, Federated Learning

Software/Cloud: React, Next.js, FastAPI, Django, Supabase, PostgreSQL, AWS, Docker, Git, Linux, REST APIs, Stripe

Certifications

AWS Certified Solutions Architect – Associate | AWS Certified Cloud Practitioner | Google Foundations of Project Management